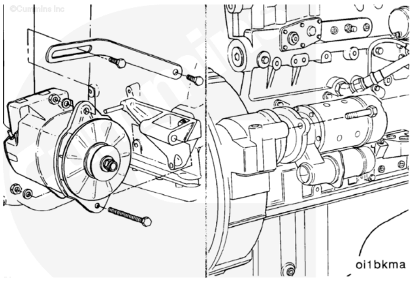
Trainings

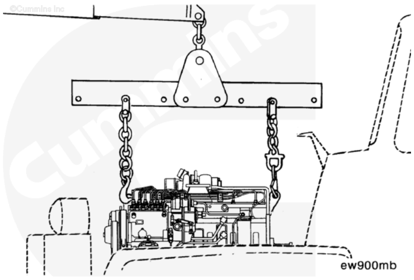
General Information

Install all accessories and brackets that were removed from the previous engine.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission safely as an assembly without causing personal injury.



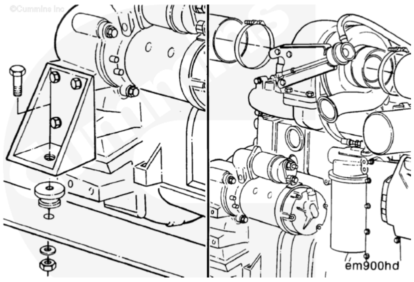
Note : If the rear engine mounts are attached to the transmission, it will probably be necessary to install the engine and the transmission as an assembly.

Use a properly rated hoist and engine lift fixture, Part Number 3163264. Attach the lift fixture to the engine-mounted lifting brackets.

Install the engine.

⚠ CAUTION ⚠

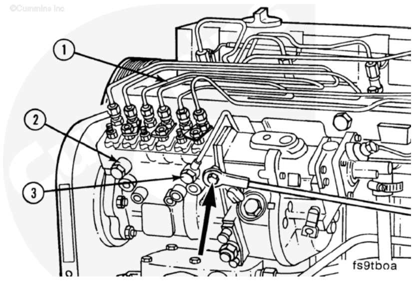
Make sure all lines, hoses, and tubes are correctly routed and fastened to prevent engine damage.



Align the engine in the chassis and tighten the engine mounting capscrews. Refer to the OEM's torque specifications.

Connect all engine and chassis-mounted accessories that were removed.

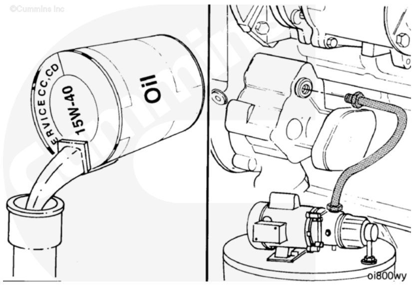
Install and adjust the throttle linkage to the injection pump control lever; refer to the OEM service manual.



Note : The total oil capacity of the standard engine, including combination full flow and bypass oil filters, is 23.6 liters [6.25 gal].

Fill the engine with clean engine lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html).

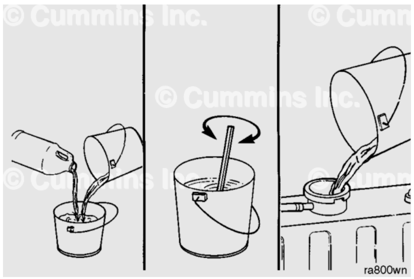
Note : The engine lubricating oil system must be pressurized before starting the engine.



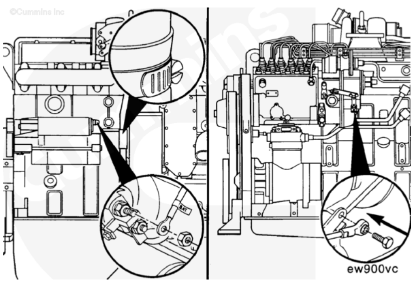
Fill the cooling system with 50-percent water, 50-percent ethylene glycol-base antifreeze, and proper amount of DCA corrosion protection. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Note : The total coolant capacity (engine only) is 9.9 liters [2.25 gal] for air-aftercooled engines, and 10.9 liters [2.85 gal] for water-aftercooled engines.

Refer to the OEM's specifications for radiator and system capacity.

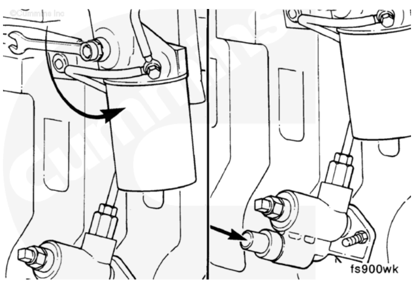


Make a final inspection to make sure all hoses, wires, linkages, and components have been correctly installed and tightened.



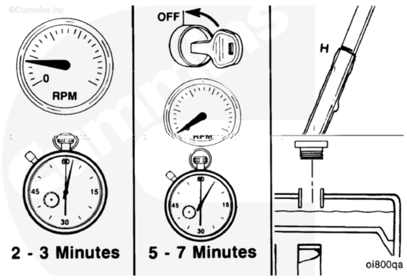
Prime the low-pressure fuel system.

1. Open the bleed screw
2. Operate the plunger on the fuel transfer pump until the fuel flowing from the fitting is free of air.
3. Close the bleed screw.



⚠ WARNING ⚠

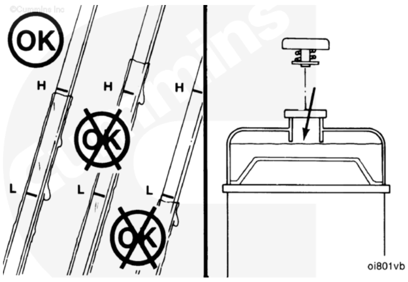
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Operate the engine at low idle for 2 to 3 minutes.

Shut off the engine and wait 5 to 7 minutes for the oil to drain to the oil pan. Check the oil and coolant levels.

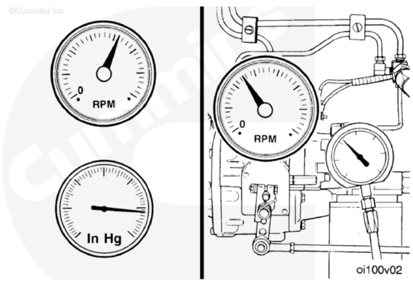
Fill the engine to the correct oil and coolant levels if necessary. Refer to Procedures 007-037 (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html) and 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).



Operate the engine at 1000 to 1200 rpm for 8 to 10 minutes to check for correct engine operation, unusual noises, and coolant, fuel, or lubricating oil leaks.

Repair all leaks and component problems as needed. Refer to the appropriate sections.

Refer to the following procedures for the Engine Run-in and Testing Procedures.



014-002 (/qs3/pubsys2/xml/en/procedures/100/100-014-002.html)
014-003 (/qs3/pubsys2/xml/en/procedures/41/41-014-003.html)
014-004 (/qs3/pubsys2/xml/en/procedures/41/41-014-004.html)